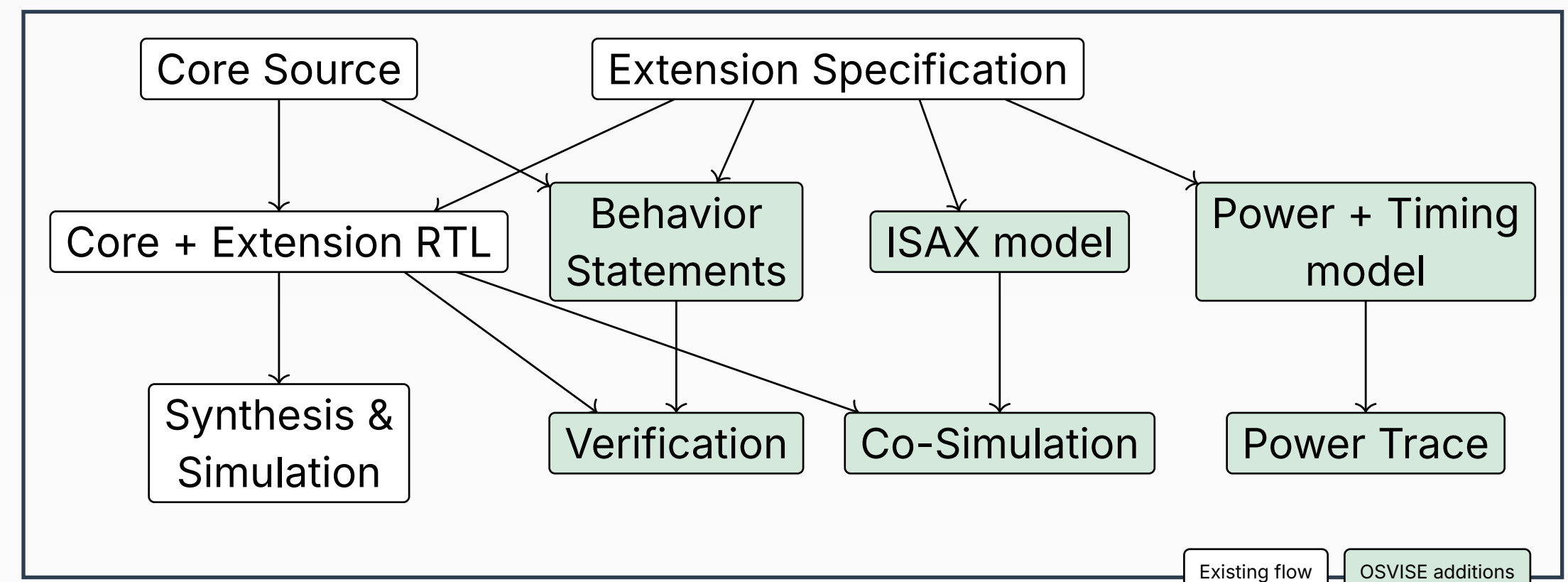


Overview

OSVISE is motivated by the need for open-source tools for verification of instruction set extensions. RISC-V enables frictionless instruction set extensions. This requires an ecosystem which supports fast evaluation, verification, and investigation of nonfunctional properties.

The OSVISE project focuses on closing the gaps in the open-source EDA ecosystem when working on Instruction Set Extensions. Central to this tool stacks are: **CIRCT, Yosys, Verilator, ETISS, CoreDSL**.



Simulation Framework

- Extending Verilator UVM capabilities
- UVM testbench for instruction extensions with tracing

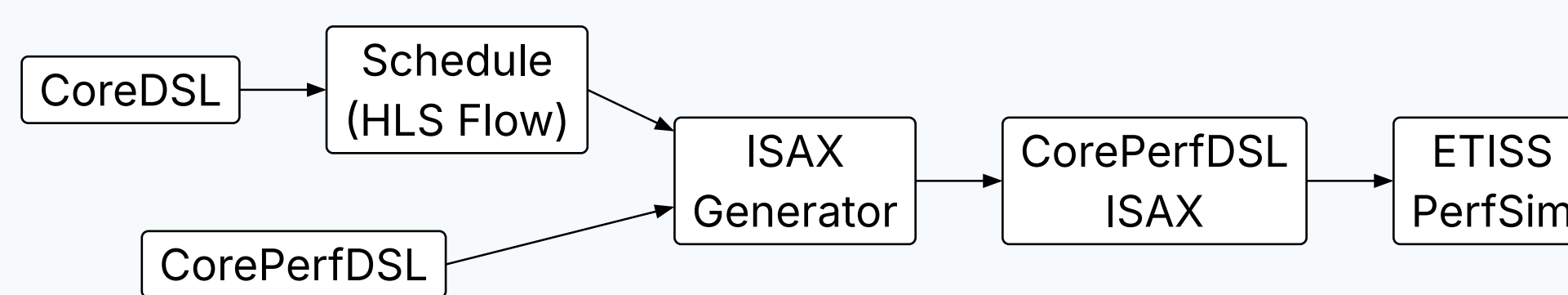
Status:

- Verilator: randomization and constraint handling
- Equivalent verification results for example extension between commercial tools and Verilator
- Successful RISC-V CPU firmware simulation using extended Verilator

Performance Modeling

CorePerfDSL Generator:

- Reuses ISAX generator flow output
- Integrates into ETISS simulators for precise performance estimates

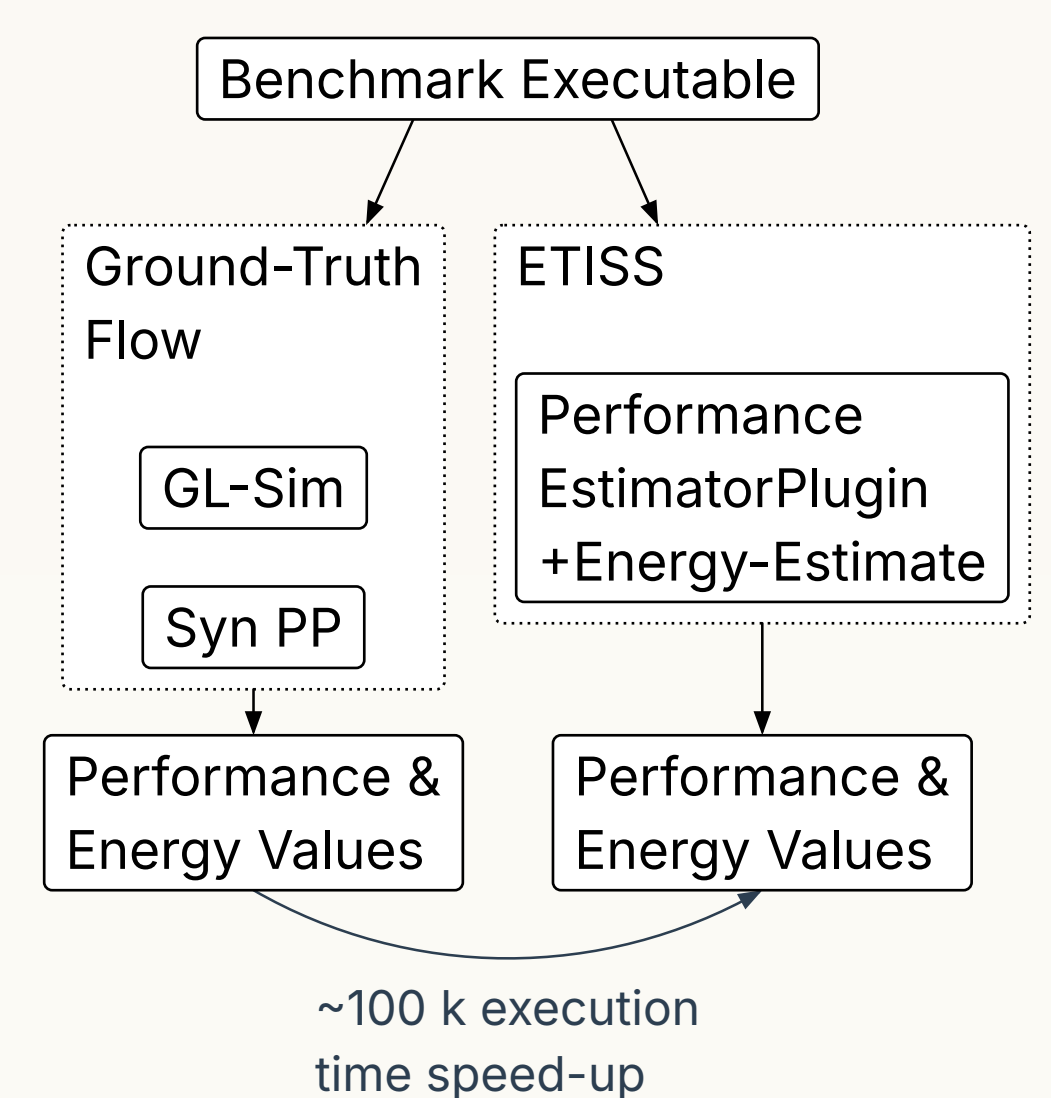


- Status:**
- Specification completed
 - Prototype implemented

Nonfunctional Verification

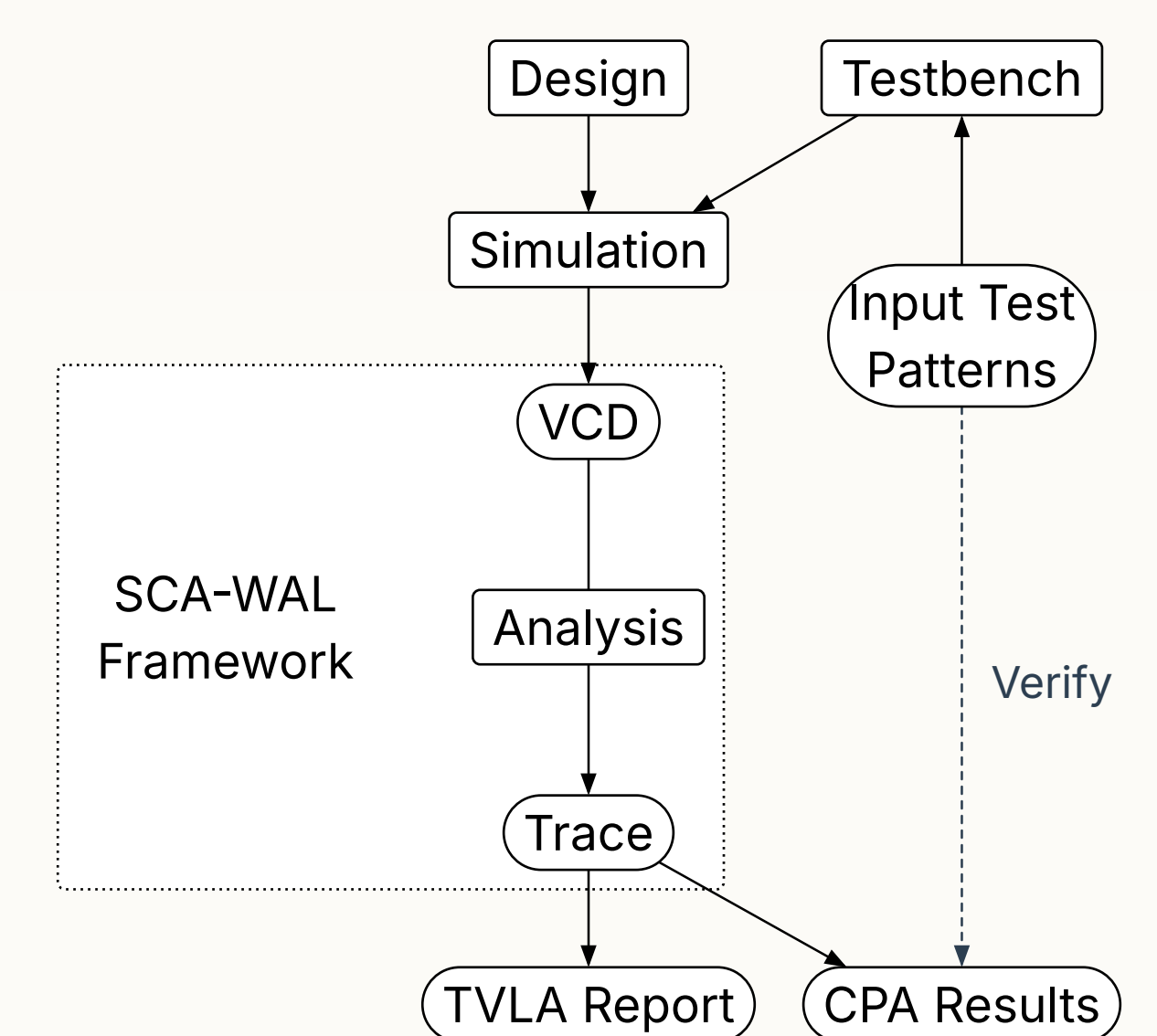
Extendable Translating Instruction Set Simulator (ETISS) with Performance Estimation:

- Demonstrates high-level simulation feasibility using Co-Simulation
- Enables rapid performance and energy evaluation
- Estimates performance and energy of generated instruction streams using the core's average power



Power Estimation:

- Power estimation based on switching activity extracted from VCDs
- Waveform parsing using the open-source Waveform Analysis Language (WAL)
- Uses Liberty files for hardware parameters and netlists for gate-level analysis



Status:

- Demonstration for RISC-V dotprod ISAX
- CorePerfDSL Generator for ISAX
- Power estimation and modeling methods survey
- Open-source analytical power estimation model for RTL and gate-level

Instruction Stream

Instruction Stream Generator:

- Architectural-aware instruction stream for checks across multiple instructions
- Specification of constraints to describe architecture
- ISAX specified in CoreDSL

- Status:**
- Specification completed
 - Prototype implemented

Open-Source Impact

- Development is done in the open
- Enables outside collaboration
- Allows for direct feedback

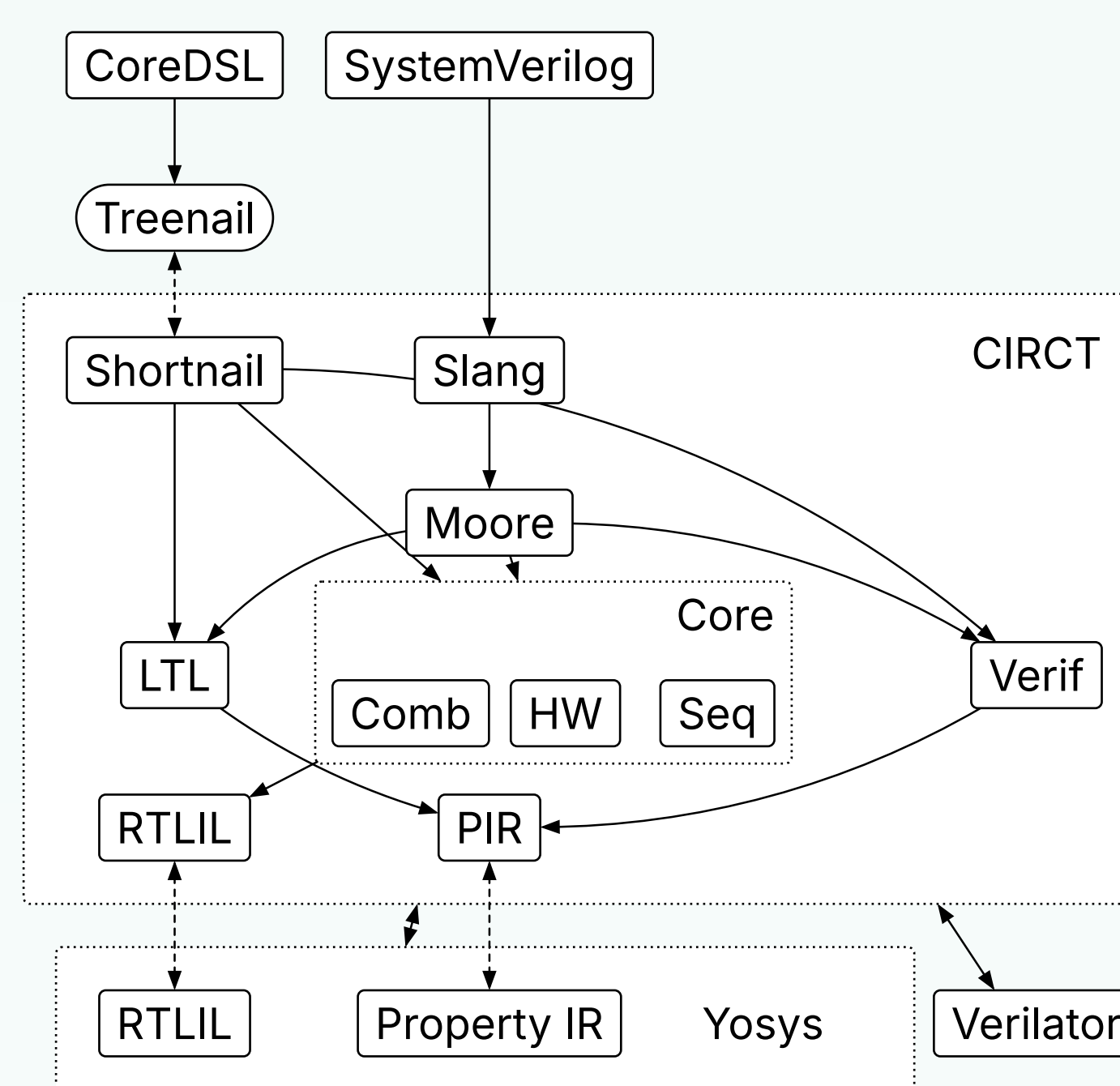
- Public access to new tools
 - tum-ei-eda/isaac-perf-gen
 - Minres/CRIG
 - planvtech/PlanV_Verilator_Feature_Tests
 - hm-aemy/coredsl-to-sva
 - hm-aemy/Shortnail
 - Extends: esa-tu-darmstadt/Shortnail
 - hm-aemy/chosys
 - YosysHQ/property-ir
 - tum-ei-eda/PerformanceSimulation_workspace
 - iti-luebeck/sca-wal
 - Extends: ics-jku/wal
- Contributions to existing tools
 - llvm/circt
 - tum-ei-eda/etiss
 - YosysHQ/yosys
 - verilator/verilator

Evolving the open-source EDA ecosystem through ongoing contributions, ensuring immediate impact and continuous improvement.

Functional Verification

SystemVerilog Assertions:

- Hardware verification is crucial, yet open-source tooling remains limited
- OSVISE extends open-source support for concurrent assertions
- CIRCT acts as a central hub, mapping diverse inputs to central core dialects via specific conversions



Instruction Extension

- Extract verification statements from instruction behavior
 - Create checker circuits from verification statements with transformations in Yosys

New Design Flows: Using CIRCT as an interchange framework enables novel methodologies. For example, *Chosys* provides a *Chisel*-to-Yosys path without requiring SystemVerilog.

- Status:**
- Prototype for CoreDSL to SVA conversion
 - CIRCT to Yosys via RTLIL demonstrated
 - Yosys Property IR defined

